

Quantitative Investing

PROBLEM SET 3

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Contents. Part I tests the CAPM on the 49 value-weighted industry portfolios. Part III contains the three standalone AI exercises and the capstone, a cross-sectional test of U.S. sector ETFs. Part II, which repeats the tests on the 25 size and book-to-market portfolios and examines Roll’s critique using in-sample and out-of-sample tangency portfolios, and AI Exercise 3 are appended at the end, reproduced from their own reports and retaining their original pagination.

1 PART I: 49 INDUSTRY PORTFOLIOS

Data and conventions. The 49 value-weighted industry portfolios, their average firm size and book-to-market ratios, and the market proxy are taken from `Problem_Set3_2026.xlsx`. All figures are in percent per month. Missing observations are screened at -99 , which removes the -99.99 and -999 codes; the assignment’s gloss of “any return < -1 ” assumes decimal returns, and applied literally to percent data would discard 19,602 valid observations against 2,868 genuine missing values. Industry returns are total returns while the market proxy is reported as $R_M - R_f$, so we subtract the monthly riskless rate from each industry return and work in excess returns throughout. Six industries have no data before the late 1960s, and all 49 series are populated only from July 1969; we therefore report the balanced July 1969 – June 2026 panel ($T = 684$, $N = 49$) as our primary specification and the full July 1926 – June 2026 sample ($T = 1200$, N from 40 to 49) as a robustness check. Eleven negative book-equity observations are masked, since the logarithm is undefined for them. The sample mean of $R_M - R_f$ is 0.6116 on the balanced sample and 0.6952 on the full sample.

1.1 Question (a): CAPM predictions for γ_0 and γ_M

In the cross-sectional regression $R_i = \gamma_0 + \gamma_M \beta_{iM} + \eta_i$ with $\beta_{iM} = \text{cov}(R_i, R_M) / \sigma^2(R_M)$, each observation is a portfolio, so γ_0 is the expected return on a portfolio with zero market beta and γ_M is the expected return per unit of beta.

Sharpe/Lintner. With unrestricted borrowing and lending at the riskless rate, $E[R_i] = R_f + \beta_{iM}(E[R_M] - R_f)$, so

$$\gamma_0 = R_f, \quad \gamma_M = E[R_M] - R_f.$$

Both are exact point predictions. A portfolio with $\beta_{iM} = 0$ bears no systematic risk and must earn the riskless rate, which fixes the intercept. The market has $\beta_{iM} = 1$, so the line must pass through $(1, E[R_M])$, which fixes the slope at the market risk premium.

Black. Without riskless borrowing and lending, R_f is replaced by the expected return on the zero-beta portfolio R_z , the minimum-variance portfolio uncorrelated with the market, giving $E[R_i] = E[R_z] + \beta_{iM}(E[R_M] - E[R_z])$ and

$$\gamma_0 = E[R_z], \quad \gamma_M = E[R_M] - E[R_z].$$

Because $E[R_z]$ is not observable, this version predicts no specific intercept. It restricts only $\gamma_M > 0$ and $\gamma_0 < E[R_M]$, and is therefore weaker and harder to reject: Sharpe/Lintner is Black plus the

restriction $E[R_z] = R_f$. Both imply $\gamma_0 + \gamma_M = E[R_M]$, so they differ only in where the security market line is anchored.

Because all tests use excess returns, the Sharpe/Lintner null becomes $\gamma_0 = 0$ with γ_M equal to the average market excess return, and Black predicts $\gamma_0 = E[R_z] - R_f$, unrestricted in sign but generally positive.

1.2 Question (b): Fama-MacBeth estimation

Each industry's beta is estimated once by OLS over the full sample and held fixed thereafter; the estimates range from 0.516 (Utilities) to 1.514 (Software), with mean 1.039. In each of the 684 months we then estimate $R_{it} = \gamma_{0t} + \gamma_{Mt}b_{iM} + \eta_{it}$ across the 49 industries. The monthly slopes are highly volatile, with a standard deviation of 7.13 against a mean of 0.035, and are positive in only 48.4% of months. Finally we average the monthly estimates, using

$$\hat{\gamma}_j = \frac{1}{T} \sum_{t=1}^T \hat{\gamma}_{jt}, \quad s(\hat{\gamma}_j) = \frac{\sigma(\hat{\gamma}_{jt})}{\sqrt{T}},$$

so that the standard error is the time-series standard deviation of the monthly estimates divided by $\sqrt{T}$, not the cross-sectional OLS standard error from any single month.

Table 1: Fama-MacBeth estimates, 49 industry portfolios, percent per month. The final column gives the value each coefficient should take if the CAPM holds, with $\gamma_0 = 0$ under Sharpe/Lintner in excess-return space.

	Estimate	Std. Error	<i>t</i>	<i>p</i>	CAPM
<i>Balanced, July 1969 – June 2026, T = 684</i>					
γ_0	0.6119	0.2191	2.79	0.005	0
γ_M	0.0352	0.2725	0.13	0.897	0.6116
<i>Full, July 1926 – June 2026, T = 1200</i>					
γ_0	0.5160	0.1714	3.01	0.003	0
γ_M	0.2490	0.2315	1.08	0.282	0.6952

Can we reject mean-variance efficiency? Yes. Both restrictions fail: the intercept is significantly positive where it should be zero, and the slope is indistinguishable from zero and significantly below the realized market premium ($t = -2.12$). Taken together these describe a single pattern, the flat security market line, which is too high at $\beta = 0$ and too shallow thereafter. That pattern licenses the stronger conclusion, because the exact linear relation between $E[R_i]$ and β_{iM} holds if and only if the proxy is mean-variance efficient with respect to the test assets. The positive intercept alone would be consistent with Black, which permits $E[R_z] > R_f$, but that reading requires the zero-beta rate to absorb essentially the entire market premium while beta goes unpriced. The full sample gives the same conclusion with slightly less extreme flattening.

AI integration: code generation. We first asked an LLM to explain three aspects of the procedure, limiting it to three lines each, so the omissions below reflect our prompt rather than its knowledge.

- **Fixed betas.** It answered that betas need time-series variation to estimate and that short rolling windows would let noise correlate mechanically with the same month's return. This matches our understanding. We would add that fixed betas also make the monthly $\hat{\gamma}_{Mt}$ comparable across months.

- **Standard errors.** It correctly identified cross-sectional correlation in residuals as the source of the difference and predicted correctly that the Fama-MacBeth errors would be larger. It did not note that the two methods give identical point estimates in a balanced panel.
- **Errors-in-variables.** It correctly described attenuation of the slope and the offsetting upward bias in the intercept, without adding that Fama-MacBeth does not fix this, since the same estimated betas enter every month.

We then asked it to implement the procedure and ran its script ourselves.

- The delivered code was correct; the errors it reported were from earlier drafts, namely a 36-month rolling beta, a missing $\sqrt{T}$ divisor, and an assumed fixed cross-section of 49. Two corrections were ours: it ran the full rather than the balanced sample, and it described the missing codes as “−99.99 or −999” when only −99.99 occurs here.
- Yes. It divides the time-series standard deviation by $\sqrt{T}$, and our independent run reproduces its standard errors exactly. It contrasted this against a pooled panel OLS rather than against the cross-sectional regression the assignment describes.
- Yes on substance, but this assessment is the model’s own, so we treat it as a claim and verified it by re-running the script against our data.

1.3 Question (c): Cross-sectional regression on time-averaged returns

Table 2: Fama-MacBeth compared with OLS on time-averaged returns, balanced sample. The point estimates are identical; only the standard errors differ.

	γ_0		γ_M	
	Estimate	Std. Error	Estimate	Std. Error
(b) Fama-MacBeth	0.6119	0.2191	0.0352	0.2725
(c) OLS on averages	0.6119	0.1154	0.0352	0.1086

Are the estimates different? No. Because the betas are fixed and the panel is balanced, every monthly regression uses the same regressor, so each monthly coefficient is a weighted average of that month’s returns with constant weights. Averaging those coefficients over time is the same calculation as regressing the time-averaged returns on beta. The estimates coincide exactly only in the balanced panel; in the full sample the number of industries varies across months, so the two methods weight months differently.

Are the standard errors different, and which method is superior? The Fama-MacBeth standard errors are roughly twice as large, because the two methods measure different sources of uncertainty. The cross-sectional regression treats the average returns as fixed and uses only the dispersion of the 49 residuals, which averaging over time has already made small. Fama-MacBeth instead uses the month-to-month variation in the estimated coefficients, which is large. Fama-MacBeth is superior: since the point estimates are the same, only the standard errors matter, and the cross-sectional regression assumes residuals are independent across industries. That assumption is not reasonable here, because industries share a common market shock. On the full sample this matters for the conclusion, as the cross-sectional regression makes $\hat{\gamma}_M$ appear significant while Fama-MacBeth does not.

1.4 Question (d): Average return against estimated beta

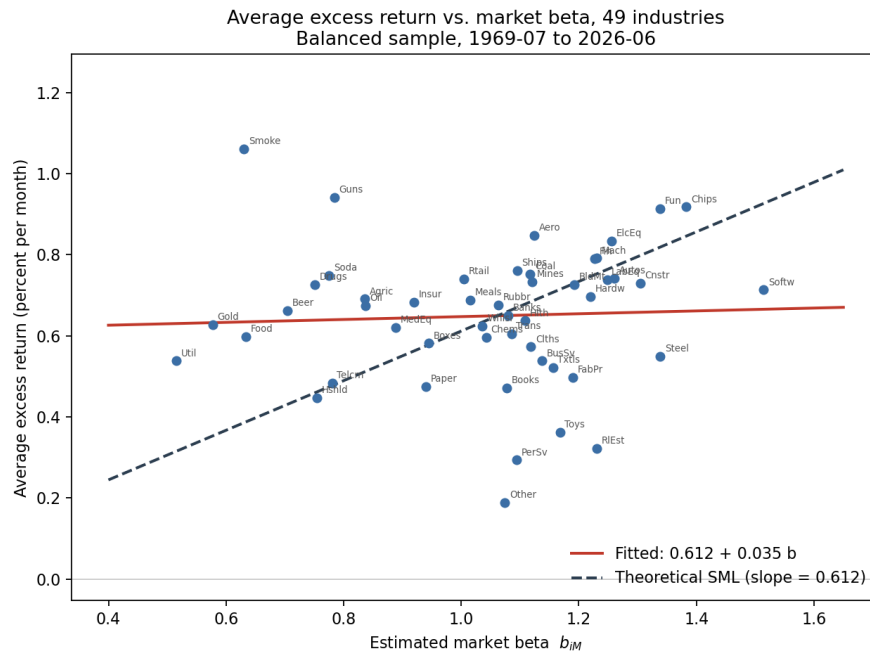


Figure 1: Average excess return against estimated beta, 49 industry portfolios, July 1969 – June 2026. Solid line: fitted cross-sectional regression, $\bar{R}_i = 0.612 + 0.035 b_{iM}$ with $R^2 = 0.002$. Dashed line: theoretical SML through the origin with slope 0.612.

Does the plot resemble a positive relationship? Barely. The fitted line is almost flat, and beta explains essentially none of the cross-sectional variation in average returns. Low-beta industries such as Utilities, Gold and Food earn about as much as high-beta industries such as Software, Chips and Steel. The vertical scatter at any given beta is much wider than the variation explained by beta itself, which is visible around $b_{iM} \approx 1.1$, where average returns range from Other near the bottom of the chart to Ships and Coal near the top.

What should the plot look like? Under the CAPM the points should lie along an upward-sloping straight line. In excess-return space that line passes through the origin, because a zero-beta portfolio should earn no premium, and its slope should equal the average market excess return. Deviations should be small and unsystematic, because beta is supposed to be the only characteristic that is priced. Instead the fitted line is far too high at $\beta = 0$ and far too shallow, so it crosses the theoretical line near $\beta = 1$ and diverges from it in both directions. This is the flat security market line of part (b) shown graphically.

AI integration: SML prediction. Asked what the scatter should look like and what the most common deviation is, the LLM answered that the points should lie on a line through the origin with slope equal to the market premium, that the estimated line is typically too flat with a positive intercept, and that this reflects leverage constraints under Black’s zero-beta CAPM, citing the Frazzini-Pedersen “betting against beta” result, and omitted risk factors such as size and value.

- (i) The empirical SML is much flatter than the theoretical one. The model predicted the direction correctly but understated the degree, describing a line that still slopes upward, whereas our slope is statistically indistinguishable from zero.

- (ii) It mentions all three, and its account of the leverage mechanism is correct. It does not mention errors-in-variables in the estimated betas, which produces the same flattening for purely statistical reasons, or Roll's critique, under which a flat line may indicate a poor market proxy rather than a failed model. The response also referred to our results, so it was not predicting blind.
- (iii) No. Beta explains almost none of the cross-sectional variation, and industries with very different betas earn similar average returns. Utilities and Software have the lowest and highest betas in the sample, yet their average returns differ by little.

1.5 Question (e): CAPM predictions for γ_{size} and $\gamma_{B/M}$

In the extended regression

$$R_i = \gamma_0 + \gamma_M \beta_{iM} + \gamma_{size} \ln(\text{size}) + \gamma_{B/M} \ln(\text{BE/ME}) + \eta_i,$$

the CAPM predicts $\gamma_{size} = \gamma_{B/M} = 0$. Expected return depends only on covariance with the market, so beta is a sufficient statistic and no other characteristic should have explanatory power once beta is included. Size and book-to-market are firm characteristics rather than measures of systematic risk, so they should be priced only if they proxy for beta, and beta is already controlled for. The predictions for γ_0 and γ_M are unchanged from part (a).

This regression tests the Fama-French (1992) critique. A significantly negative γ_{size} would indicate a size premium and a significantly positive $\gamma_{B/M}$ a value premium. Either result rejects the CAPM, regardless of the estimate of γ_M .

1.6 Question (f): Extended Fama-MacBeth estimation

Using the same fixed betas, we estimate each month

$$R_{it} = \gamma_{0t} + \gamma_{Mt} b_{iM} + \gamma_{size,t} \ln(\text{size}_{t-1}) + \gamma_{B/M,t} \ln(\text{BE/ME}_{t-1}) + \eta_{it},$$

with size lagged one month and book-to-market taken from the previous year and held constant within the year.

Why must the characteristics be lagged? They must be known to investors before the return is realized. Size is measured by market capitalization, which moves with the return itself, so contemporaneous size would place part of the month's return on both sides of the regression and create a spurious relationship. Book-to-market has the same problem through its market value denominator, and annual accounting data is not public until several months after the fiscal year ends. Lagging makes the regression a test of whether information available at the start of the month predicts the return earned over it.

Table 3: Extended Fama-MacBeth estimates, balanced sample, July 1969 – June 2026 ($T = 684$), percent per month. The average cross-sectional R^2 rises from 0.09 in part (b) to 0.18.

	Estimate	Std. Error	t	CAPM
γ_0	0.4045	0.4988	0.81	0
γ_M	0.0328	0.2595	0.13	0.6116
γ_{size}	0.0428	0.0497	0.86	0
$\gamma_{B/M}$	0.1568	0.0805	1.95	0

Can we reject mean-variance efficiency? Yes, for the same reason as in part (b): the slope on beta is still indistinguishable from zero and significantly below the average market excess return ($t = -2.23$). Adding the characteristics does not restore a risk-return relationship.

The characteristics themselves give only weak evidence against the CAPM. The coefficient on book-to-market is positive and marginally significant, which is consistent with a value premium, while the coefficient on size is small and insignificant. This is not surprising for industry portfolios, which are not sorted on these characteristics and therefore show limited cross-sectional variation in them; the 25 portfolios in Part II provide a sharper test. The intercept is no longer significant once the characteristics are included, which suggests the zero-beta return found in part (b) is partly absorbed by book-to-market rather than being a separate effect. Over the full sample both characteristics are insignificant and the intercept is significantly positive, with the rejection again coming from the slope on beta ($t = -2.83$).

AI integration: interpreting size and B/M premia. The prompt specified in the assignment asserts significant premia on both characteristics, which our results do not show, so we ran it as written and then supplied our actual table, to test whether the model would confirm the premise it was given. Asked the specified question it identified the rejection at once, noting that “formally, you’re rejecting the joint restriction $\gamma_{size} = \gamma_{B/M} = 0$, which the CAPM requires,” and offered three interpretations: that the characteristics proxy for priced risks a single market index does not capture, following Fama and French; that “it’s the characteristics themselves, not the covariance structure, that predict returns,” following Daniel and Titman, which points to mispricing; and that a significant loading is “also consistent with the market proxy simply being wrong,” following Roll.

- (i) Yes. It identified the rejection immediately and stated the restriction the CAPM imposes.
- (ii) Yes, and more fully than required, contrasting the risk-based reading with the mispricing and invalid-proxy alternatives.
- (iii) Only partly. It noted that the characteristics are known before the return is earned, but did not make the point that lagging solves the econometric problem without settling whether the characteristics represent genuine risk factors.

Given our actual table it declined to claim a size effect and identified the beta slope as the main source of rejection, though it misstated $t = 1.95$ as roughly 5% one-sided and 10% two-sided when the correct values are 2.6% and 5.2%. Asking the model to interpret real output proved a better test than asking it to explain a result stated in the prompt.

2 PART II: 25 SIZE AND BE/ME PORTFOLIOS

Sample. 1969-07 to 2026-06, $T = 684$, $N = 25$. Estimates and standard errors: percentage points per month. Mean market excess return: 0.6116.

2.1 Question 1(b)

Table 4: Fama–MacBeth estimates (t -statistics test zero). Average monthly cross-sectional R-squared: 0.2425. Slope restriction $E(\gamma_{M,t} - \text{MktRF}_t) = 0$: mean difference = -1.1873 , SE = 0.3916, $t = -3.0316$, $p = 0.0025$.

	Estimate	FMB SE	t -stat	p -value
gamma_0	1.3554	0.3998	3.390	0.0007
gamma_M	-0.5758	0.4290	-1.342	0.1800

Using fixed full-period betas and 684 monthly cross-sectional regressions, we obtain $\bar{\gamma}_0 = 1.3554$ (FMB SE = 0.3998, $t = 3.390$) and $\bar{\gamma}_M = -0.5758$ (FMB SE = 0.4290, $t = -1.342$), in percent per month. Standard errors use $\text{sd}(\hat{\gamma}_{j,t})/\sqrt{T}$.

The significantly positive intercept rejects the zero-intercept restriction. The slope also falls below the average market excess return of 0.6116%; the difference has $t = -3.0316$ and $p = 0.0025$. These results reject the Sharpe–Lintner CAPM restrictions and provide evidence against the proxy being the tangency portfolio for these assets. The insignificant slope relative to zero is not, by itself, grounds for rejection.

2.2 Question 1(c)

Table 5: Fama–MacBeth and single cross-sectional regression estimates compared. Cross-sectional R-squared: 0.1987. $N = 25$.

	FMB est.	FMB SE	FMB t -stat	OLS est.	OLS SE	OLS t -stat
gamma_0	1.3554	0.3998	3.390	1.3554	0.2619	5.1759
gamma_M	−0.5758	0.4290	−1.342	−0.5758	0.2411	−2.3884

The single cross-sectional regression gives the same coefficients as Fama–MacBeth: $\gamma_0 = 1.3554$ and $\gamma_M = -0.5758$, with $R^2 = 0.1987$ and $N = 25$. Equality follows because the panel is balanced and the regressors are fixed: averaging monthly OLS coefficients is equivalent to regressing average returns on the same betas.

The OLS standard errors, 0.2619 and 0.2411, are smaller than the FMB standard errors, 0.3998 and 0.4290. OLS measures dispersion around the average-return regression, whereas FMB measures variation in monthly coefficient estimates. FMB is preferable for inference on average premia here because it incorporates that time-series variation and accommodates contemporaneous correlation across portfolio returns. It does not eliminate error from estimated betas.

2.3 Question 1(d)

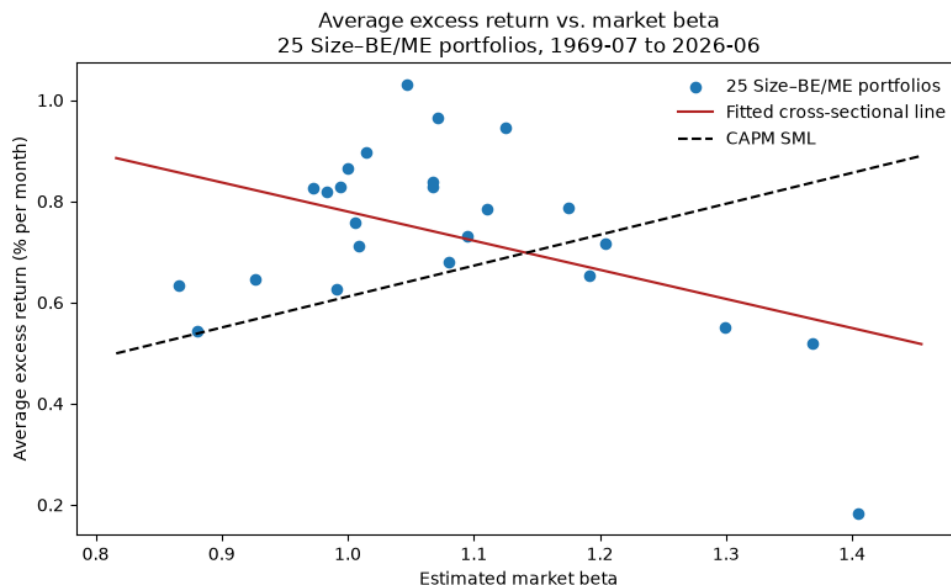


Figure 2: Average excess return against estimated market beta, 25 size-BE/ME portfolios. Fitted line: $1.3554 + (-0.5758) \times \text{beta}$. CAPM SML: $0.6116 \times \text{beta}$. Cross-sectional R-squared: 0.1987.

The plot shows a negative fitted relationship:

$$R_i^e = 1.3554 - 0.5758 \hat{\beta}_{iM}.$$

The CAPM instead predicts a line through the origin with positive slope 0.6116. The portfolios are widely scattered around the fitted line, with $R^2 = 0.1987$, so market beta explains little of their average-return variation. The plot is inconsistent with the predicted SML, although visual evidence alone is not a formal statistical test.

2.4 Question 1(f)

Table 6: Extended Fama–MacBeth estimates (t -statistics test zero). Average monthly R-squared, beta only: 0.2425; extended: 0.5288. Slope restriction $E(\gamma_{M,t} - \text{MktRF}_t) = 0$: mean difference = -0.8761 , SE = 0.2629, $t = -3.3320$, $p = 0.0009$.

	Estimate	FMB SE	t -stat	p -value
gamma_0	1.1539	0.3392	3.4017	0.0007
gamma_M	-0.2646	0.3134	-0.8442	0.3989
gamma_size	-0.0194	0.0302	-0.6428	0.5206
gamma_BM	0.1573	0.0747	2.1055	0.0356

With lagged characteristics included, the estimates are $\gamma_0 = 1.1539$ ($t = 3.4017$), $\gamma_M = -0.2646$ ($t = -0.8442$), $\gamma_{size} = -0.0194$ ($t = -0.6428$), and $\gamma_{B/M} = 0.1573$ ($t = 2.1055$). Book-to-market has a positive premium significant at 5%, while size is insignificant. Average monthly cross-sectional R^2 rises from 0.2425 to 0.5288.

The positive book-to-market premium contradicts the CAPM prediction that characteristics have no additional explanatory power after controlling for beta. The intercept remains significantly positive, and the beta premium remains below the market premium ($t = -3.3320$, $p = 0.0009$). Thus, adding characteristics does not restore the Sharpe–Lintner CAPM restrictions.

Size is lagged one month and BE/ME uses the previous year’s value, held constant within each year. This follows the assignment’s information-timing convention and avoids using contemporaneous characteristics, though actual accounting-data availability would require publication dates.

2.5 Question 2

2(a). Using the 684 monthly observations from July 1969 to June 2026, we calculate the sample mean vector $\hat{\mu}$ and covariance matrix $\hat{\Sigma}$ of the 25 portfolios’ excess returns. The tangency weights are

$$w_T = \frac{\hat{\Sigma}^{-1} \hat{\mu}}{\mathbf{1}' \hat{\Sigma}^{-1} \hat{\mu}}.$$

The weights sum to one, with short selling allowed.

Table 7: In-sample tangency weights and tangency betas. Weights are reported as fractions of portfolio value. Size and BE/ME groups increase from 1 to 5. Sum of weights: 1.000000. Gross exposure (sum of absolute weights): 13.3859. Mean excess return: 2.5204% per month. Monthly volatility: 6.0461%. Monthly Sharpe ratio: 0.4169. Weighted average beta: 1.000000.

Portfolio	Weight	Tang. beta	Portfolio	Weight	Tang. beta
Size1_BM1	-1.9858	0.0725	Size3_BM4	-0.2411	0.3253
Size1_BM2	0.6508	0.2841	Size3_BM5	0.2578	0.3834
Size1_BM3	-0.0034	0.2903	Size4_BM1	1.1460	0.2595
Size1_BM4	0.9189	0.3563	Size4_BM2	-0.6837	0.2694
Size1_BM5	0.9936	0.4092	Size4_BM3	-0.0040	0.3006
Size2_BM1	-0.2076	0.2061	Size4_BM4	0.4626	0.3274
Size2_BM2	0.8054	0.3119	Size4_BM5	-0.1819	0.3330
Size2_BM3	0.2286	0.3289	Size5_BM1	0.7176	0.2484
Size2_BM4	-0.0409	0.3435	Size5_BM2	0.0421	0.2559
Size2_BM5	-0.4743	0.3749	Size5_BM3	0.5150	0.2519
Size3_BM1	-0.6186	0.2183	Size5_BM4	-1.0900	0.2154
Size3_BM2	0.0531	0.3109	Size5_BM5	0.4014	0.3288
Size3_BM3	-0.6615	0.2819			

Gross exposure,

$$\sum_i |w_i|,$$

is 13.3859, indicating substantial offsetting long and short positions.

2(b) Returns, betas, and cross-sectional results. We construct the tangency portfolio's monthly excess return using the same fixed weights throughout:

$$R_{T,t}^e = \sum_{i=1}^{25} w_{T,i} R_{i,t}^e.$$

Its average excess return is 2.5204% per month, monthly volatility is 6.0461%, and the monthly Sharpe ratio is 0.4169.

We estimate each portfolio's beta by regressing its excess returns on $R_{T,t}^e$, including an intercept. Regressing average excess returns on these new betas gives

$$R_i^e \approx 0 + 2.5204 \hat{\beta}_{i,T}, \quad R^2 = 1, \quad N = 25.$$

The estimated intercept is effectively zero, while the slope equals the tangency portfolio's average excess return. All 25 portfolios lie essentially on the theoretical SML, so the fitted and theoretical lines overlap. The OLS standard errors are effectively zero; the resulting t -statistics are not economically meaningful because the fit is mechanically exact.

Table 8: Cross-sectional regression using the in-sample tangency portfolio as the market proxy. Cross-sectional R-squared: 1.000000. $N = 25$. Mean tangency excess return: 2.5204% per month. Maximum absolute regression residual: 9.659×10^{-15} . Fitted line: $0.0000 + (2.5204) \times \text{beta}$. CAPM SML: $2.5204 \times \text{beta}$.

	Estimate	OLS SE	t -stat
gamma_0	4.83988×10^{-15}	4.34718×10^{-15}	1.11334
gamma_M	2.52036	1.45191×10^{-14}	1.7359×10^{14}

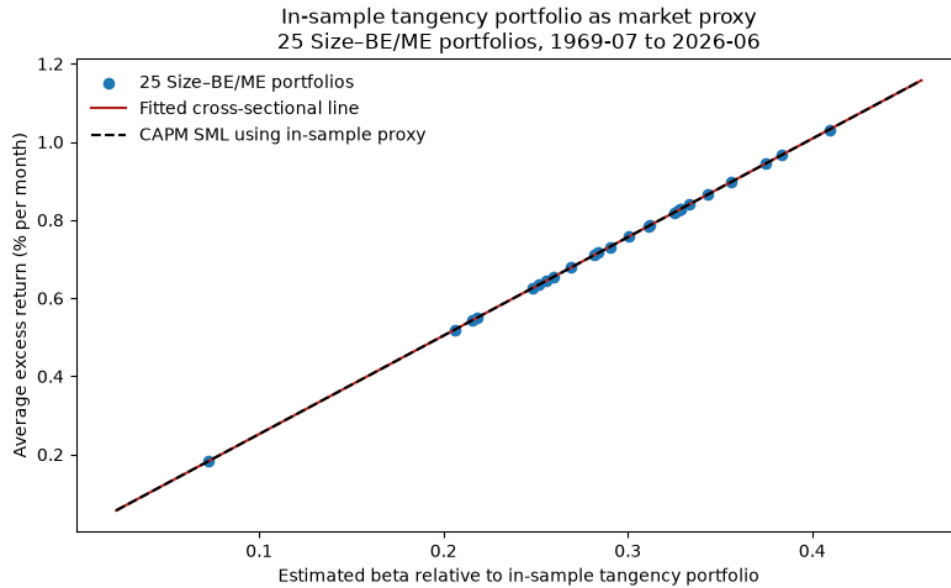


Figure 3: Average excess return against beta estimated against the in-sample tangency portfolio. Fitted line: $0.0000 + (2.5204) \times \text{beta}$. CAPM SML: $2.5204 \times \text{beta}$.

2(c). This perfect fit does not independently support the CAPM. The tangency portfolio is constructed to be mean–variance efficient using the same assets and observations used in the regression. Its weights satisfy

$$w_T \propto \hat{\Sigma}^{-1} \hat{\mu},$$

which implies

$$\hat{\mu}_i = \hat{\beta}_{i,T} \hat{\mu}_T.$$

Thus, the zero intercept, matching slope, and perfect fit follow from portfolio mathematics.

This illustrates Roll’s critique: CAPM tests depend on whether the chosen market proxy is mean–variance efficient. The true market portfolio includes all wealth and is not directly observable. Rejecting the CAPM with a particular proxy could therefore reflect a poor proxy, while obtaining a perfect fit with a deliberately constructed tangency portfolio does not establish that the true market portfolio is efficient.

2.6 Question 3

3(a). Sample 1 contains odd-numbered months in even years and even-numbered months in odd years, giving 342 observations. Using only these observations, we estimate the mean vector and covariance matrix of excess returns and calculate

$$w_1 = \frac{\hat{\Sigma}_1^{-1} \hat{\mu}_1}{\mathbf{1}' \hat{\Sigma}_1^{-1} \hat{\mu}_1}.$$

The weights sum to 1, with gross exposure $\sum_i |w_{1,i}| = 15.4800$ and individual weights ranging from -1.6671 to 1.4736 . We hold these weights fixed and apply them to the 342 months in Sample 2. These returns are out of sample because none of the evaluation observations were used to estimate the weights.

3(b). Sample 2 contains even-numbered months in even years and odd-numbered months in odd years, also giving 342 observations. We estimate a separate tangency portfolio using only Sample 2's mean vector and covariance matrix, with the same normalization.

These weights sum to 1, have gross exposure of 17.2082, and range from -2.4853 to 1.3680 . We then apply them, unchanged, to Sample 1 returns. This reverses the estimation and evaluation samples, ensuring that neither half is evaluated using weights estimated from itself. The checks confirm that the samples do not overlap and together cover all 684 months.

Table 9: Out-of-sample tangency weights. Sample 1: 342 months; Sample 2: 342 months; total 684 months, no overlap and complete coverage.

Estimated from	Applied to	Weight sum	Gross exposure	Min weight	Max weight
Sample 1	Sample 2	1.0	15.4800	-1.6671	1.4736
Sample 2	Sample 1	1.0	17.2082	-2.4853	1.3680

3(c). We combine Sample 1 returns generated with w_2 and Sample 2 returns generated with w_1 , then sort them chronologically. The resulting OOS tangency series covers July 1969–June 2026 and has a mean excess return of 1.9754% per month, monthly volatility of 7.0300%, and monthly Sharpe ratio of 0.2810.

We use this combined excess-return series as the market proxy and estimate each portfolio's beta through a full-period time-series regression with an intercept. The cross-sectional regression of average excess returns on these betas is

$$R_i^e = 0.0684 + 3.2851 \hat{\beta}_{i,\text{OOS}} + \hat{\eta}_i.$$

Table 10: Cross-sectional regression using the out-of-sample tangency portfolio as the market proxy. Returns are measured in percent per month. The regression has $R^2 = 0.9753$ and $N = 25$. Mean OOS market excess return: 1.9754% per month.

Coefficient	Estimate	OLS standard error	t -statistic
γ_0	0.0684	0.0228	2.9930
γ_M	3.2851	0.1091	30.1058

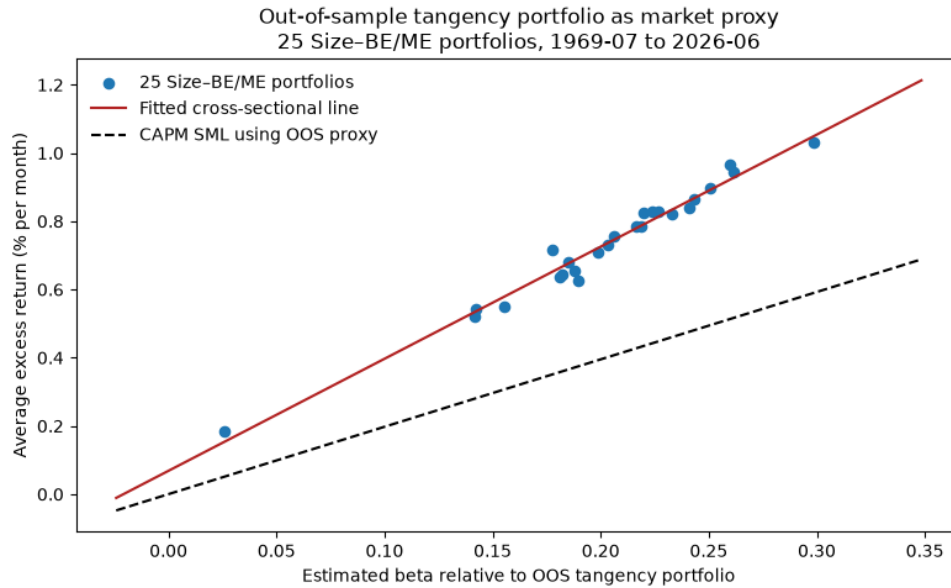


Figure 4: Average excess return against beta estimated against the out-of-sample tangency portfolio. Fitted line: $0.0684 + (3.2851) \times \text{beta}$. CAPM SML: $1.9754 \times \text{beta}$.

The plot shows a strong positive relationship: the portfolios cluster closely around the fitted line, which explains 97.53% of the cross-sectional variation in average excess returns. However, the fitted line lies above the theoretical SML, whose intercept is zero and slope is 1.9754. Thus, a close fit to the estimated line does not establish that the CAPM restrictions hold. The reported slope t -statistic tests equality to zero, not equality to 1.9754.

Each month's return is excluded from the sample used to estimate its portfolio weights. Because the two samples are interleaved in time, this is a held-out-sample exercise rather than a strictly forward-looking trading backtest.

2.7 Question 4

Table 11: In-sample and out-of-sample tangency proxies compared.

Statistic	Q2: In-sample	Q3: Out-of-sample
γ_0 (SE, t)	4.84×10^{-15} (4.35×10^{-15} , $t = 1.11$)	0.0684 (0.0228, $t = 2.99$)
γ_M (SE, t)	2.5204 (1.45×10^{-14} , $t = 1.74 \times 10^{14}$)	3.2851 (0.1091, $t = 30.11$)
Proxy mean excess return (CAPM slope)	2.5204	1.9754
$\gamma_M \div$ proxy premium	1.00	1.66
Cross-sectional R^2	1.000000	0.9753
Max absolute residual	9.66×10^{-15}	not reported
Proxy Sharpe ratio (monthly)	0.4169	0.2810

1. Comparing the results.

- γ_0 : In Q2 the intercept is zero to machine precision. The $t = 1.11$ is rounding noise, not a meaningful test. In Q3 the intercept is 0.0684% per month, about 0.82% per year, with $t = 2.99$, indicating a statistically significant common pricing error across the 25 portfolios.

- γ_M : In Q2 the slope equals the proxy's premium exactly, 2.5204. In Q3 the slope is 3.2851, about 1.66 times the proxy premium of 1.9754. The raw slopes should not be compared directly across Q2 and Q3 because each set of betas is measured against a different proxy. What matters for the CAPM is whether the slope equals that proxy's own premium. It does in Q2 but not in Q3.
- Standard errors: Q2's standard errors are essentially zero because the regression residuals are essentially zero, so the enormous slope t -statistic is not economically informative. Q3's t -statistics use ordinary cross-sectional OLS standard errors, which do not account for beta-estimation error or cross-portfolio dependence. Also, $t = 30.11$ tests $\gamma_M = 0$, not the CAPM restriction $\gamma_M = 1.9754$.
- SML shape: In Q2 the fitted line, $R_i^e = 0.0000 + 2.5204\beta_i$, and the theoretical SML are the same line. In Q3 the fitted line, $R_i^e = 0.0684 + 3.2851\beta_i$, has both a higher intercept and a steeper slope than the theoretical SML, $R_i^e = 1.9754\beta_i$. Thus, for positive betas, the fitted Q3 line lies above the theoretical SML.
- Which is cleaner? Q2 fits the CAPM restrictions mechanically: the intercept is zero, the slope equals the proxy premium, and $R^2 = 1$. Both Q2 and Q3 fit much better than the value-weighted market proxy in Part I, where $\gamma_0 = 1.3554$, $\gamma_M = -0.5758$, and $R^2 = 0.1987$.

2. Why Q2 and Q3 differ, and the link to Roll (1977). The in-sample proxy is efficient by construction. Using excess returns, the tangency weights are

$$w_T = \frac{\hat{\Sigma}^{-1}\hat{\mu}}{\mathbf{1}'\hat{\Sigma}^{-1}\hat{\mu}}.$$

Let

$$c = \mathbf{1}'\hat{\Sigma}^{-1}\hat{\mu}.$$

Then the result follows in four steps:

1. Rearranging the tangency-weight formula gives $\hat{\Sigma}w_T = \hat{\mu}/c$.
2. $\hat{\Sigma}w_T$ is the vector of sample covariances between each portfolio and the tangency portfolio: $\widehat{\text{Cov}}(R_i, R_T)$.
3. The tangency portfolio's variance is $w_T'\hat{\Sigma}w_T = \hat{\mu}_T/c$.
4. Therefore,

$$\hat{\beta}_i = \frac{\widehat{\text{Cov}}(R_i, R_T)}{\widehat{\text{Var}}(R_T)} = \frac{\hat{\mu}_i}{\hat{\mu}_T}, \quad \text{which implies} \quad \hat{\mu}_i = \hat{\beta}_i\hat{\mu}_T.$$

This identity holds for every portfolio.

Q2 estimates the weights using the same 684 months and the same 25 portfolios that are subsequently used in the CAPM test. Because the OLS betas are also based on the same sample covariances and variances, the beta-pricing identity must hold essentially exactly. The maximum residual of 9.66×10^{-15} confirms this.

Why the CAPM can look successful even when the market is mismeasured. Roll's critique is that an empirical CAPM test is fundamentally a test of whether the chosen market proxy is mean-variance efficient. The true CAPM market portfolio contains all relevant wealth and cannot be directly observed.

The results illustrate the problem. With the observable value-weighted market proxy in Part I, the CAPM performs poorly. With a proxy constructed directly from the test assets in Q2, the fit becomes perfect. Nothing about the true market changed; only the proxy changed.

Therefore, Q2's apparent success mainly shows that the proxy was constructed to be mean-variance efficient relative to these 25 portfolios in this sample. A perfect fit does not independently establish that the true market portfolio is efficient, while rejection using another proxy could reflect proxy error rather than failure of the CAPM itself.

Why the out-of-sample version is more informative. In Q3, each month's proxy return uses tangency weights estimated from the opposite half of the data. Therefore, the means and covariances used to construct the proxy are no longer the same moments used to evaluate it. The exact in-sample identity above no longer has to hold.

Estimation error therefore becomes important:

- Inverting the 25×25 covariance matrix in $\hat{\Sigma}^{-1}\hat{\mu}$ can magnify estimation noise.
- This produces large long and short positions. Gross exposure is 15.48 and 17.21 in the two estimation samples, with individual weights ranging roughly from -2.4853 to 1.4736 .
- Weights optimized using one half of the data may partly fit noise specific to that half. When applied to the other half, performance deteriorates. The proxy mean falls from 2.5204% to 1.9754% per month, volatility rises from 6.0461% to 7.0300%, and the monthly Sharpe ratio falls from 0.4169 to 0.2810.
- Because the OOS proxy is no longer guaranteed to lie on the evaluation-sample mean-variance frontier, pricing errors appear: $\gamma_0 > 0$, the fitted slope differs from the proxy premium, and $R^2 < 1$.

The remaining high R^2 of 0.9753 indicates that the beta-return relation remains strong out of sample, even though the exact in-sample CAPM identity disappears.

Limits of the Q3 test:

- The two samples alternate months and therefore span the same broad economic periods, so this is a relatively mild out-of-sample test.
- The second-pass average returns and betas are calculated using the combined sample, so the design is not equivalent to a completely separate future holdout period.
- The OOS proxy is still constructed from the same 25 test portfolios rather than from the true aggregate market portfolio. Therefore, Q3 makes the test more informative but does not eliminate Roll's critique.

3. Evaluating the LLM explanation. 1. Did it predict a steeper SML and lower pricing errors? Partly.

The LLM correctly predicted that an in-sample tangency proxy would generate small pricing errors and a very strong beta-return relationship. It actually understated the strength of the result: it called the exercise “almost tautological,” while Q2 produces an essentially exact fit with $R^2 = 1.000000$.

However, it did not explicitly predict a steeper SML. In fact, “steeper” is not the key criterion. Q3's fitted slope, 3.2851, is numerically steeper than Q2's 2.5204. The important CAPM restriction is instead whether the fitted slope equals the proxy's own mean excess return. That equality holds exactly in Q2 but fails in Q3.

2. Did it explain why the portfolio is efficient by construction? Yes. This is the strongest part of the LLM explanation.

It correctly argues that the tangency first-order condition implies

$$\mu - r_f \mathbf{1} \propto \Sigma w_T,$$

and recognizes that Σw_T is the vector of covariances between each test asset and the tangency portfolio. This leads directly to beta pricing.

There are two minor precision issues:

- It skips the intermediate step needed to determine the proportionality constant. Premultiplying by w_T' gives the connection between the constant, the tangency portfolio premium, and its variance.
- It writes the result using population expectations such as $E[R_i]$, whereas the exact identity here is an *in-sample* identity involving sample moments, such as $\hat{\mu}_i$. This distinction explains why the Q2 fit is exact in the observed sample but need not hold out of sample.

3. Did it anticipate why the out-of-sample results could deteriorate? In general terms, yes. The LLM correctly explained that moments in the evaluation sample need not equal the moments used to estimate the weights, so mean-variance efficiency is no longer guaranteed.

However, its explanation was less complete about the empirical pattern observed in Q3. It did not emphasize:

- estimation error amplified by $\hat{\Sigma}^{-1}$;
- the resulting extreme leveraged weights;
- the lower out-of-sample Sharpe ratio;
- the positive intercept and slope exceeding the proxy premium;
- the fact that R^2 nevertheless remains high at 0.9753;
- the exact two-way sample-splitting procedure used in Q3;
- or the fact that the OOS proxy is still not the true market portfolio, so Roll's critique remains relevant.

Overall, the LLM's explanation is conceptually correct and explains the exact in-sample fit in Q2 well. It is less complete in explaining the magnitude and specific pattern of the deterioration observed in Q3.

3 PART III, AI EXERCISE 1: STRESS-TESTING LLM KNOWLEDGE OF FAMA-MACBETH ECONOMETRICS

LLM used: ChatGPT. The three questions were asked one at a time in a single conversation.

3.1 Question 1: Shanken (1992) Correction

Prompt. What is the Shanken (1992) correction and when should it be applied to Fama-MacBeth standard errors?

Summary of the LLM's Answer. The LLM explained that the second-pass Fama-MacBeth regression uses estimated betas, but conventional Fama-MacBeth standard errors treat them as known, so they overstate the precision of the estimated risk premia. It gave Shanken's adjusted covariance matrix, $V_{Shanken} = (1 + c)\Omega + \Sigma_f^*$ with $c = \lambda' \Sigma_f^{-1} \lambda$, compared with the conventional $V_{FM} = \Omega + \Sigma_f^*$, so corrected standard errors are larger and t -statistics smaller. It said the correction should be applied when the second-pass regressors are estimated betas, but generally not when they are observed characteristics such as size or book-to-market, and that it matters more when betas are noisy or the time series used to estimate them is short. It also noted that Shanken's result assumes conditional homoskedasticity and that the correction is different from Newey-West, which addresses serial correlation and heteroskedasticity.

Evaluation. The answer is mostly correct. The LLM correctly identifies the source of the problem (estimated betas treated as known), the direction of the error in conventional standard errors (too small), when the correction applies and when it does not, and the difference from Newey-West. Its general formula is consistent with the assignment's answer: with a single market factor, $c = \lambda^2/\sigma_M^2$, so the correction inflates the conventional variance by $(1 + \lambda^2/\sigma_M^2)$. However, it never reduces the formula to this one-factor case, so the key result is left implicit. Its statement that the correction matters more when the beta-estimation sample is short is imprecise: the size of the correction relative to the conventional standard error depends on c , the squared Sharpe ratio of the factor, not on the sample length. It also does not mention that c is small for a monthly market factor, so the correction is usually modest in practice.

My Correction. In the one-factor case, $c = \lambda^2/\sigma_M^2$, so the Shanken correction inflates the conventional Fama-MacBeth variance (apart from the factor-variance term) by $1 + \lambda^2/\sigma_M^2$, and because this term is the squared Sharpe ratio of the market rather than a function of sample length, it is usually small for monthly market data.

3.2 Question 2: Errors-in-Variables

Prompt. Does the Fama-MacBeth procedure fully solve the errors-in-variables problem caused by using estimated betas as regressors? If not, what bias remains?

Summary of the LLM's Answer. The LLM said Fama-MacBeth does not fully solve the errors-in-variables problem: the same noisy estimated betas are the regressors in every monthly cross-sectional regression, so averaging the monthly risk-premium estimates does not remove the measurement error. With one factor and classical measurement error, the estimated premium is attenuated toward zero by the factor $\text{Var}(\beta)/[\text{Var}(\beta) + \text{Var}(e)]$; with several factors, the estimates are jointly distorted and not every coefficient is necessarily biased toward zero. It added that the Shanken correction adjusts the standard errors but does not remove the bias in the coefficients, and that the bias shrinks as the first-stage beta estimates become more precise.

Evaluation. The answer is correct but incomplete. The LLM correctly explains why averaging does not remove the measurement error, identifies attenuation bias toward zero in the one-factor case, gives the attenuation factor, notes that the multi-factor case cannot always be signed, and usefully distinguishes the bias (which Shanken does not fix) from the standard errors (which it does). However, it does not state that the bias persists even in large samples, meaning that adding more assets does not reduce it. It also does not mention that attenuation of the slope biases the intercept upward, producing a flatter estimated security market line, or that grouping stocks into portfolios is the standard way to reduce beta measurement error.

My Correction. Because the same noisy betas enter every monthly regression, the estimated market premium γ_M is attenuated toward zero, and with a positive premium the intercept is biased upward, even when the number of assets is large, so the bias can only be reduced by estimating betas more precisely, for example by using portfolios.

3.3 Question 3: Serial Correlation in Monthly Gamma Estimates

Prompt. In a FMB regression where the same betas are used every month, are the monthly gamma estimates serially uncorrelated? Does this matter for the standard error calculation?

Summary of the LLM’s Answer. The LLM said that using the same betas every month does not by itself make the monthly gamma estimates serially correlated. Because the betas are fixed, each month’s estimate is $\hat{\gamma}_t = (\hat{B}'\hat{B})^{-1}\hat{B}'R_t$, so the gammas are serially correlated only if returns or regression disturbances are correlated over time, which can happen in real data. The usual standard error $s(\hat{\gamma}_t)/\sqrt{T}$ assumes no serial correlation; if the gammas are positively autocorrelated, it is usually too small and the t -statistics too large, and Newey-West/HAC standard errors should be used. It again distinguished Newey-West (serial correlation) from Shanken (estimated betas).

Evaluation. The answer is mostly correct. It identifies the mechanism behind the assignment’s answer: because the betas are fixed, each $\hat{\gamma}_t$ is a fixed linear combination of that month’s returns, so the gammas inherit whatever autocorrelation the returns have. It also correctly explains why this matters, the direction of the resulting error in the standard errors, and the Newey-West remedy. Its framing (fixed betas do not by themselves cause serial correlation) differs in emphasis from the assignment’s (fixed betas are why the gammas inherit return autocorrelation), but the two are consistent. However, it does not address magnitude: it presents Newey-West as the remedy without noting that autocorrelation in monthly returns, and therefore in the gammas, is typically mild.

My Correction. Because the betas are fixed, each monthly gamma is a fixed-weight portfolio return, so the gammas are serially correlated only to the extent that returns are, which is typically mild for monthly data, so the usual $s(\hat{\gamma}_t)/\sqrt{T}$ standard error is usually adequate, with Newey-West as a robustness check.

4 PART III, AI EXERCISE 2: PREDICTING CROSS-SECTIONAL DEVIATIONS

We asked an LLM which of the 49 industries it expected to fall above, below and near the security market line, and recorded its answer before comparing it with our results from Question (d).

(a) The LLM’s predictions.

- *Above the line:* Tobacco, Utilities, Beer & Liquor / Food.
- *Below the line:* Coal / Mining, Computers / Electronic Equipment / Semiconductors, Autos / Steel.
- *Near the line:* Wholesale / Retail, Transportation, Financials / Business Services.

Its reasoning was the low-beta anomaly. Defensive industries have low beta but still earn competitive returns, so they plot above the line, while cyclical industries have high beta without the returns to match. It added a sin-stock premium for tobacco and a glamour effect for technology. It stated that these were predictions from documented patterns, not calculations from the data.

(b) What the plot shows. Deviations are measured from the theoretical SML in Figure 1, as $\bar{R}_i - \bar{R}_M^e b_{iM}$.

Table 12: Largest deviations from the theoretical security market line, balanced sample, percent per month. A positive value means the industry earned more than its beta implies.

Above		Below		Near	
Smoke	+0.675	Other	−0.468	BldMt	−0.003
Guns	+0.461	RlEst	−0.429	Boxes	+0.004
Soda	+0.274	PerSv	−0.376	Telcm	+0.006

(c) How accurate were the predictions? The above-line predictions were accurate. All three industries plot above the line, and Tobacco is the largest positive deviation in the sample. The near-line predictions were also reasonable: Wholesale, Transportation and Financials all sit close to the line, though none is among the three closest.

The below-line predictions were mostly wrong. Coal and Mining plot slightly above the line rather than below it. Of the technology group, only Software is clearly below, while Chips and Electronic Equipment are above. Only Steel was placed correctly, and none of the three largest negative deviations was anticipated.

This asymmetry is explained by how the model reasoned. It used beta alone, predicting that low-beta industries plot above the line and high-beta industries plot below. Because the estimated line is so flat, the first half of this is almost automatic, since nearly any low-beta industry lands above the theoretical SML. That is why the above-line predictions worked. The below-line predictions needed more, because high beta is not sufficient: Chips and Electronic Equipment have high betas but earned enough to stay above the line. The industries that actually fall furthest below have betas near the sample average and unusually low returns, which reflects industry-specific weakness rather than a risk story the model could infer.

Its caveat about technology was partly right. It predicted technology would fall below the line but noted that recent data might reverse this. The reversal is what we observe for Chips and Electronic Equipment, though not for Software.

5 PART III, AI EXERCISE 3: LITERATURE SYNTHESIS AND HISTORICAL CONTEXT

5.1 3(a)

Table 13: These are the values reported in Part I(b). Estimates and standard errors are in percent per month; the table's t -statistics test against zero. The reported mean market excess returns are 0.6116% and 0.6952% per month, respectively. For the main sample, the report gives $t = -2.12$ for testing the slope against the sample market premium.

Sample	Parameter	Estimate	Standard error	t	p
July 1969–June 2026	γ_0	0.6119	0.2191	2.79	0.005
July 1969–June 2026	γ_M	0.0352	0.2725	0.13	0.897
July 1926–June 2026	γ_0	0.5160	0.1714	3.01	0.003
July 1926–June 2026	γ_M	0.2490	0.2315	1.08	0.282

My Part I(b) results weaken the favorable evidence of Fama and MacBeth (1973) and contradict the Sharpe–Lintner CAPM restrictions in my sample. For July 1969–June 2026, $\hat{\gamma}_0 = 0.6119\%$ per month (SE = 0.2191, $t = 2.79$, $p = 0.005$), significantly above the zero intercept predicted for excess returns. Meanwhile, $\hat{\gamma}_M = 0.0352\%$ per month (SE = 0.2725, $t = 0.13$, $p = 0.897$), providing no statistically significant evidence of a positive beta premium. The slope is also far below the CAPM benchmark of 0.6116%, with the reported benchmark test rejecting equality at the 5% level ($t = -2.12$). Thus, the estimated security market line is nearly flat, failing to replicate their positive beta–return relationship. The longer July 1926–June 2026 sample reinforces this pattern: the intercept remains significantly positive, while the beta premium remains insignificant. A positive excess-return intercept alone does not reject Black's zero-beta CAPM, but the weak beta–return relationship undermines the earlier favorable interpretation.

5.2 3(b)

The original LLM response correctly identifies Banz's (1981) size effect: small firms earned higher average returns than their market betas could explain. It also accurately summarizes Fama and French (1992), emphasizing the explanatory power of size and book-to-market and the weak relation between beta and average returns once beta variation is separated from size. However, it omits Rosenberg, Reid, and Lanstein (1985), an important earlier study documenting the book-to-market effect that the assignment explicitly asks about. Although its discussion of Fama and French (1993) adds useful context, it does not fill this omission. Overall, the response captures the main empirical challenges to the CAPM but gives an incomplete account of the development of the value evidence.

5.3 3(c)

The original LLM response does not acknowledge data snooping: size and book-to-market may appear important partly because researchers identified and tested these patterns using the same historical data. This concern also applies to Part I(f), since our sample overlaps with the data used to discover these effects, so it is not a fully independent test. However, we use previously documented characteristics rather than search across many predictors ourselves. Our results provide only weak evidence: size is insignificant ($t = 0.86$), while book-to-market is marginally significant ($t = 1.95$). Lagging these characteristics addresses their timing but does not eliminate data snooping. Testing on a separate period after the original studies would provide stronger independent evidence.

5.4 3(d)

LLM response to record. International evidence shows that size and value effects occur outside the United States, although their strength varies across markets and samples.

- **Direct Fama–MacBeth evidence:** Hu, Chen, Shao, and Wang (2019) find a significant size-factor premium in Chinese stocks, while market and value-factor premia are insignificant. Thus, international Fama–MacBeth tests do not uniformly support both size and value. [1]
- **Broader international evidence:** Rouwenhorst (1999) finds that small stocks outperform large stocks and value stocks outperform growth stocks in emerging markets. Fama and French (1998) find value outperformance in twelve of thirteen major markets. These studies provide complementary evidence from portfolio comparisons and factor tests; their findings should not all be described as Fama–MacBeth regression results. [2], [3]

My interpretation is that international replication reduces the likelihood that these effects are merely artifacts of searching U.S. data. It is consistent with compensation for omitted risks, but does not establish that explanation: similar investor biases and limits to arbitrage could also produce recurring patterns across countries.

Evaluation for submission. The international evidence modestly strengthens the plausibility of a risk-based interpretation by showing that size and value patterns are not confined to U.S. data. However, it strengthens the evidence that these patterns exist more clearly than it distinguishes risk from mispricing, since investor biases and limits to arbitrage may also operate internationally. The LLM appropriately acknowledges this distinction and the variation across markets. In our Part I(f), size is insignificant ($t = 0.86$) and book-to-market is only marginally significant ($t = 1.95$), so international findings cannot turn our weak evidence into proof of priced risks. They provide supporting context for the positive book-to-market coefficient, but establishing a risk explanation would require evidence linking returns to exposure to economically meaningful risks.

Sources cited in 3(d). [1] Hu, Chen, Shao, and Wang (2019). “Fama–French in China: Size and Value Factors in Chinese Stock Returns.” doi:10.1111/irfi.12177
 [2] Rouwenhorst (1999). “Local Return Factors and Turnover in Emerging Stock Markets.” doi:10.1111/0022-1082.00151
 [3] Fama and French (1998). “Value versus Growth: The International Evidence.” doi:10.1111/0022-1082.00080

6 CAPSTONE: CROSS-SECTIONAL TEST OF U.S. SECTOR ETFs

6.1 Step 1: Dataset

Of the datasets offered, this capstone uses **U.S. sector ETF returns**.

6.2 Research Question

Does market beta explain cross-sectional differences in average returns across U.S. sector ETFs?

6.3 Hypothesis

The dependent variable is the average monthly return of each U.S. sector ETF, and the main independent variable is its market beta relative to SPY.

I expect sector ETFs with higher market betas to have higher average returns. Under the CAPM, the relationship should be positive, with the coefficient on beta approximately reflecting the market risk premium. Economically, investors should require higher expected returns for bearing greater systematic market risk.

I would view the results as supportive of the CAPM if the beta coefficient is positive and statistically significant, with an intercept reasonably close to the risk-free rate. If the beta coefficient is insignificant, zero, or negative, or if the intercept differs substantially from the risk-free rate, I would conclude that the CAPM does not explain the cross-section of sector ETF returns well.

Specification clarification (added after analysis). The implemented regression uses excess returns rather than raw returns. Therefore, the Sharpe-Lintner CAPM predicts an intercept of zero, rather than the risk-free rate. The original hypothesis above is retained as written before examining the data.

6.4 Methodology

I use monthly returns for nine U.S. sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, and XLY), with SPY as the market proxy and the risk-free rate from the Kenneth French Data Library.

The sample covers February 2000 through December 2025, with 311 monthly observations.

I first calculate excess returns by subtracting the monthly risk-free rate from each ETF’s return and SPY’s return. I then estimate each ETF’s market beta using a time-series regression of ETF excess returns on SPY excess returns.

Finally, I run a cross-sectional OLS regression of average ETF excess returns on their estimated market betas. This regression provides the coefficient estimates, while Fama-MacBeth standard errors are used as the main basis for statistical inference.

To calculate the Fama-MacBeth (FMB) standard errors, I run monthly cross-sectional regressions using the same fixed betas. Each month, I regress the nine ETF excess returns on the same fixed full-sample betas, then take the time-series average of the 311 monthly intercept and slope estimates. The FMB standard error is the time-series standard deviation of the monthly estimates divided by

$\sqrt{T}$. Because the betas are the same every month, the FMB coefficient averages equal the OLS coefficients exactly; only the standard errors differ.

6.5 Results

Table 14: Cross-sectional CAPM test on nine sector ETFs. Number of sector ETFs: 9. Sample period: February 2000–December 2025. Monthly return observations (T): 311. Cross-sectional R^2 : 0.3915. Average SPY excess return: 0.0060. OLS p -values use $N - 2 = 7$ degrees of freedom; FMB p -values use $T - 1 = 310$ degrees of freedom. All coefficients and standard errors are reported in decimal units per month. For comparison with Part I, which reports returns in percent per month, the estimates should be multiplied by 100.

Variable	Coefficient	OLS SE	OLS t	OLS p	FMB SE	FMB t	FMB p
Intercept	0.0045	0.0009	4.8158	0.0019	0.0028	1.6032	0.1099
Beta	0.0020	0.0010	2.1220	0.0715	0.0037	0.5506	0.5823

Table 15: Beta premium against market premium. The test of the difference uses the monthly series $\gamma_{Mt} - \text{SPY}_t$, whose time-series standard error accounts for the sampling uncertainty of both averages and their covariance. There is no equivalent monthly series for the single cross-sectional OLS regression, so under OLS the comparison is reported descriptively only.

	Value
Estimated beta premium (γ_M)	0.0020
Average SPY excess return	0.0060
Difference ($\gamma_M - \text{SPY}$)	-0.0040
Std. error of difference (monthly series)	0.0027
t -stat of difference	-1.47
p -value of difference	0.1416

6.6 Step 4(a): Hypothesis

The estimated beta premium is positive at 0.0020 (0.20% per month), which is directionally consistent with my original hypothesis. However, it is not statistically significant at the 5% level under either method (OLS $p = 0.0715$; FMB $p = 0.5823$).

Under the CAPM, the beta premium should equal the market risk premium. The estimated beta premium of 0.20% per month is about one third of the average SPY excess return of 0.60% per month. The estimated intercept is 0.0045 (0.45% per month), whereas the Sharpe-Lintner CAPM predicts an intercept of zero for excess returns. Together, a positive intercept and a slope below the market premium are consistent with a flatter security market line than the CAPM predicts.

The statistical strength of this evidence depends on the standard errors. Using the main OLS standard errors, the intercept is significant ($p = 0.0019$). Using FMB standard errors, which account for the common monthly shocks that move all sector ETFs together, neither the intercept ($p = 0.1099$) nor the beta premium ($p = 0.5823$) is statistically significant. The gap between the beta premium and the average SPY excess return, tested with the monthly difference series, is also not statistically significant ($t = -1.47$, $p = 0.1416$).

Therefore, the point estimates suggest a flatter SML than the CAPM implies, but the FMB robustness check shows that this test does not have enough precision to reject the CAPM. The results neither strongly support my hypothesis nor provide statistically reliable evidence against the CAPM using SPY as the market proxy.

6.7 Step 4(b): Comparison with Parts I and II

My sector ETF results show a similar pattern to Parts I and II: market beta has limited power to explain cross-sectional differences in average returns. The estimated beta premium is 0.20% per month for my nine ETFs, compared with 0.035% for the 49 industry portfolios and -0.576% for the 25 size/BE-ME portfolios. None of these beta coefficients is statistically significant using Fama–MacBeth standard errors, and all three are below their respective average market excess returns of approximately 0.60–0.61% per month.

However, the statistical evidence differs. Parts I and II report significantly positive intercepts of 0.612% and 1.355% per month, while my ETF intercept of 0.45% is not significant using Fama–MacBeth standard errors ($p = 0.110$). The difference between my estimated beta premium and the average SPY excess return is also insignificant ($p = 0.142$), whereas Part II reports a significant difference from its market premium ($p = 0.0025$). Part II additionally finds a significant book-to-market premium after controlling for beta, but no significant size premium. My ETF test does not include these characteristics.

Overall, the point estimates suggest a flatter security market line across all three sets of assets, but my ETF results provide less statistically precise evidence against the CAPM. The comparison is also limited by different sample periods and market proxies: Parts I and II cover July 1969–June 2026, while my ETF sample covers February 2000–December 2025.

6.8 Step 4(c): Main Limitation

The main limitation is the small cross-section. The second-stage regression contains only nine sector ETFs, leaving only seven degrees of freedom in the OLS regression. In addition, sector ETFs are broad portfolios with relatively similar exposure to the U.S. equity market, which limits the cross-sectional variation in beta. The estimated betas range from about 0.48 (XLU) to 1.29 (XLK), and the spread comes mainly from three defensive sectors (XLP, XLU and XLV).

As a result, the test has limited statistical power. The FMB standard errors are roughly three to four times larger than the OLS standard errors, and the test based on the monthly difference series cannot statistically distinguish the estimated beta premium of 0.20% per month from the average SPY excess return of 0.60% per month ($p = 0.1416$). The OLS standard errors treat the nine ETF residuals as independent, even though sector returns are strongly correlated, so the OLS results likely overstate the precision of the estimates.

Finally, the second-stage regression uses estimated rather than true betas. First-stage estimation error can bias the beta premium toward zero, and neither the OLS nor the FMB standard errors account for this first-stage uncertainty.

6.9 Step 3: AI Workflow Log

Step 3(a): Data Loading and Cleaning. *Prompt.* I asked the LLM to generate Python code to download monthly adjusted-price data for nine U.S. sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY) and SPY from January 2000 through December 2025, calculate monthly returns, align the series to a common sample, and check for missing observations.

What the LLM got right. The code successfully downloaded the ETF data, aligned the series, calculated monthly returns, and produced a common sample from February 2000 through December 2025 with 311 monthly return observations.

Correction. The initial code checked for missing values only after dropping missing observations, which made the diagnostic uninformative. I revised the code to report missing values before and after cleaning and explicitly used `fill_method=None` when calculating percentage changes.

After the revision, the raw dataset contained no missing prices, so no observations were removed during the cleaning step.

Step 3(b): Cross-Sectional Regression. *Prompt (close paraphrase).* Please generate Python code to estimate the market beta of each of the nine U.S. sector ETFs using SPY as the market proxy. Then run a cross-sectional OLS regression of each ETF's average monthly return on its estimated beta. Report the coefficients, standard errors, t -statistics, p -values, and R -squared.

Summary of the LLM's response. The LLM generated code that (1) estimated each ETF's beta with a full-sample time-series OLS regression of its monthly return on SPY's monthly return, (2) stored each ETF's average monthly return and beta in a table, and (3) ran a cross-sectional OLS regression of average returns on beta, reporting coefficients, standard errors, t -statistics, p -values and R^2 . When run, the code produced an intercept of 0.0061, a beta coefficient of 0.0020 ($p = 0.0721$) and an R^2 of 0.3903. It followed the prompt correctly, but it used raw returns for both the ETFs and SPY rather than excess returns.

Subsequent correction. The initial AI-generated code estimated market betas and ran the cross-sectional regression using raw returns. After reviewing the CAPM specification, I revised the code to use excess returns by subtracting the monthly risk-free rate from both ETF returns and SPY returns. The final analysis uses the excess-return specification.

The revised code uses the risk-free rate from the Kenneth French Data Library. Under this specification, the Sharpe-Lintner CAPM predicts an intercept of zero.

Robustness check: Fama-MacBeth standard errors (added after the final OLS results).

An AI code review of my finished analysis pointed out that the cross-sectional OLS standard errors treat the nine ETF residuals as independent, even though sector returns are strongly correlated. It recommended reporting Fama-MacBeth (FMB) time-series standard errors, which Part I of the problem set identifies as the preferred approach.

Prompt (close paraphrase). Keep my original cross-sectional OLS regression as the main methodology. Add Fama-MacBeth standard errors only as a robustness check, using the same fixed full-sample betas and monthly ETF excess returns. Report the OLS and FMB results side by side, including coefficients, standard errors, t -statistics and p -values, and compare the estimated beta premium with the average SPY excess return. Do not add Shanken, Newey-West or other extensions.

What the LLM produced. Code that runs the cross-sectional regression of the nine ETF excess returns on the fixed betas each month, averages the 311 monthly estimates, and computes FMB standard errors as the time-series standard deviation of the monthly estimates divided by $\sqrt{T}$. It also added a check that the FMB coefficient averages equal the OLS coefficients, which must hold when the betas are fixed.

Verification. I reran the script. The OLS results were unchanged, the FMB coefficients matched the OLS coefficients exactly, and only the standard errors differed. The estimated beta premium (0.0020) is about one third of the average SPY excess return (0.0060). The gap is not statistically significant ($t = -1.47$, $p = 0.1416$).

Later correction to the gap test. The first version of this comparison divided the gap by the standard error of the beta premium alone ($t = -4.19$ with the OLS standard error and $t = -1.09$ with the FMB standard error). This ignored the sampling uncertainty of the SPY average and its covariance with the beta premium, even though the monthly beta-premium estimates and SPY excess returns are strongly correlated. I replaced it with a test based on the monthly difference series $\gamma_{Mt} - \text{SPY}_t$, whose time-series standard error (0.0027) accounts for both. There is no equivalent test for the single cross-sectional OLS regression, so the OLS comparison is now reported descriptively

only.

Step 3(c): Interpretation Assistance. The first prompt and interpretation below refer to the initial raw-return regression. A separate interpretation of the final excess-return results is documented under “Final excess-return results” below.

Prompt. I ran a cross-sectional CAPM test on nine U.S. sector ETFs using monthly data from February 2000 to December 2025. The cross-sectional regression of average monthly returns on estimated market beta produced an intercept of 0.0061 ($t = 6.47$, $p = 0.0003$), a beta coefficient of 0.0020 ($t = 2.12$, $p = 0.0721$), and an R-squared of 0.3903. Please interpret these results. Do they support my hypothesis that higher-beta sector ETFs should have higher average returns, and how strong is the evidence for the CAPM?

LLM interpretation. The beta coefficient is positive, which is consistent with the hypothesis that higher-beta sector ETFs have higher average returns. A one-unit increase in beta is associated with approximately a 0.20 percentage point increase in average monthly return. However, the beta coefficient has a p -value of 0.072, so it is not statistically significant at the conventional 5% level, although it is significant at the 10% level. The R-squared of 0.390 indicates that market beta explains about 39% of the cross-sectional variation in average sector ETF returns. Overall, the results show a positive relationship between beta and average return, but the statistical evidence for the CAPM is limited.

My evaluation: what the LLM got right. It correctly read the sign and size of the beta coefficient (about 0.20 percentage points of monthly return per unit of beta), correctly noted that $p = 0.072$ is significant at the 10% level but not at the 5% level, and reached a reasonably cautious conclusion that the evidence for the CAPM is limited.

What it missed. It did not discuss the intercept, even though it was the most significant estimate ($t = 6.47$). In a raw-return regression, the CAPM predicts an intercept equal to the risk-free rate, so the intercept should have been compared with the average risk-free rate. It also did not compare the beta coefficient with the average market excess return, which is the slope the CAPM predicts, and it did not mention the small cross-section (nine ETFs, seven degrees of freedom) or that the first-pass betas are estimated.

What it got wrong or interpreted misleadingly.

- It took the OLS p -values at face value. These standard errors treat the nine ETF residuals as independent, even though sector returns are strongly correlated, so they overstate precision. With Fama-MacBeth standard errors, the beta coefficient has $p = 0.58$ rather than 0.07.
- It judged the CAPM mainly by whether the beta coefficient was positive and significant. The CAPM makes a sharper prediction: the slope should equal the market risk premium, and the intercept should equal the risk-free rate. A positive slope well below the market premium points to a flatter security market line rather than support for the CAPM.
- It described the R^2 of 0.39 as beta explaining 39% of the cross-sectional variation, without noting that it is estimated from only nine observations.

Some of these gaps partly reflect my prompt, which did not give the risk-free rate or the average market return.

Table 16: Specifications compared.

Specification	Intercept	Beta coefficient	Beta p -value	R^2
Original (raw returns, OLS SE)	0.0061 ($p = 0.0003$)	0.0020	0.0721	0.3903
Final (excess returns, OLS SE)	0.0045 ($p = 0.0019$)	0.0020	0.0715	0.3915
Final (excess returns, FMB SE)	0.0045 ($p = 0.1099$)	0.0020	0.5823	—

Re-assessment with the final results. After switching to excess returns, the beta coefficient was essentially unchanged, and the intercept fell to 0.0045. Under OLS standard errors, the intercept remained significant, which I first read as evidence against the CAPM. The Fama-MacBeth robustness check changed that conclusion: neither the intercept ($p = 0.1099$) nor the beta premium ($p = 0.5823$) is significant, and the gap between the beta premium (0.20% per month) and the average SPY excess return (0.60% per month) is not significant either ($p = 0.1416$). The point estimates suggest a flatter security market line than the CAPM predicts, but the test is not precise enough to reject the CAPM.

The comparison with the market premium and the concern about OLS standard errors came from the later AI code review described in Step 3(b), not from the original LLM interpretation or my first evaluation.

Final excess-return results: new LLM interpretation. I supplied the final results table, including the OLS and FMB standard errors, the cross-sectional R^2 , the number of ETFs, the average monthly SPY excess return, and the difference test, and asked whether the results support my hypothesis, what they imply about the CAPM, and how the conclusions differ when using OLS versus Fama-MacBeth standard errors.

Summary of the LLM's answer. ChatGPT explained that the estimated beta coefficient of 0.0020 is positive, which is consistent with my original hypothesis, but it is not statistically significant at the 5% level under either OLS or Fama-MacBeth standard errors.

It noted that the estimated intercept of 0.45% and beta premium of 0.20% per month suggest a flatter security market line than the CAPM predicts. The OLS intercept is statistically significant, but neither the intercept nor the beta coefficient is significant using Fama-MacBeth standard errors. The difference between the estimated beta premium and the average SPY excess return is also insignificant ($p = 0.1416$).

ChatGPT concluded that the point estimates suggest a flat SML, but the FMB results do not provide strong enough evidence to reject the CAPM's individual restrictions. It also mentioned the small cross-section, estimated-beta uncertainty, potential serial correlation and the absence of a formal joint test.

My evaluation. The LLM correctly distinguishes OLS from Fama-MacBeth inference and compares the estimated beta premium with the market premium rather than just testing whether the coefficient is different from zero. It also correctly notes that failing to reject the individual CAPM restrictions does not prove the CAPM holds, especially without a formal joint test.

However, its discussion of the limitations could be more specific. The approximate 95% FMB confidence interval for the beta premium ranges from -0.53% to 0.93% per month. It includes both zero and the 0.60% market premium, showing how imprecise this test is. The LLM mentioned estimated-beta uncertainty but did not explain that measurement error can bias the slope toward zero and the intercept upward, potentially producing a flatter SML. It also did not discuss Roll's critique: SPY is only a market proxy and overlaps heavily with the sectors being tested, so the results cannot establish whether the true market portfolio is efficient.

I did not identify a major numerical error in its interpretation. Its main weakness was leaving these implications underdeveloped.

Code. The commented, runnable script used to load the data, run the regressions and generate the results table is submitted separately as `capstone_analysis.py` (Capstone Deliverable 2).

6.10 Step 5: AI Workflow Reflection

Using an LLM let me move quickly from a research design to working Python code for downloading, cleaning and analyzing the ETF data, so I spent more of my time checking the output than writing code from scratch. That checking mattered: I revised the missing-value diagnostics after noticing that the initial code only checked for missing observations after they had already been dropped, and I changed the regression from raw returns to excess returns so that it matched the Sharpe-Lintner CAPM. A later AI code review then pointed out that the cross-sectional OLS standard errors ignore the strong correlation among sector returns, so I added Fama-MacBeth standard errors as a robustness check, which showed that neither the intercept nor the beta premium is statistically significant. AI therefore helped at two different stages, first by generating code quickly and later by questioning my statistical inference, but each improvement still depended on my verifying the specification and results against the CAPM and the methods from Part I. Next time, I would specify the CAPM form, excess returns, the missing-value diagnostics and the appropriate standard errors in my initial prompts, and I would compare the beta premium with the market premium before interpreting the results.